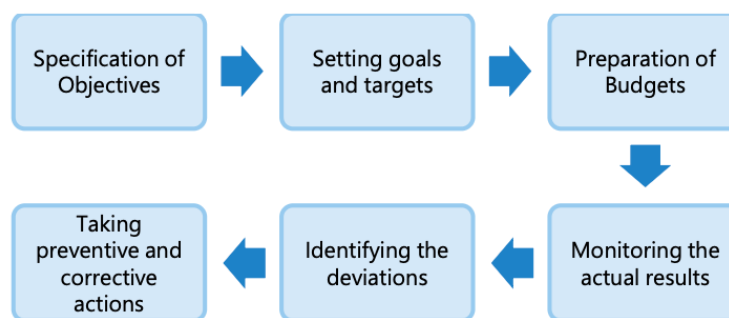


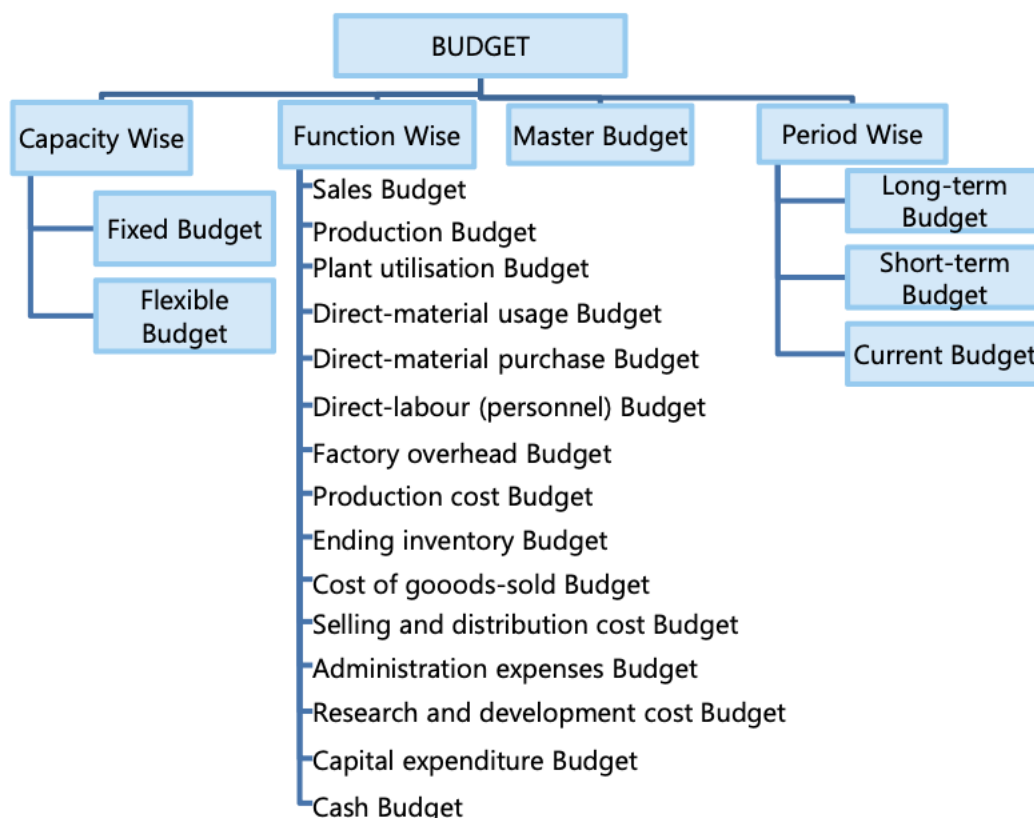
## BUDGET & BUDGETARY CONTROL

**1. Budget** – A budget is an instrument of management used as an aid in the planning, programming and control of business activity. In other words, budget is a blue-print of the projected plan of action expressed in quantitative terms for a specified period of time.

**2. Budgetary Control** – It is the system of management control and accounting in which all the operations are forecasted and planned in advance to the extent possible and the actual results compared with the forecasted and planned results.



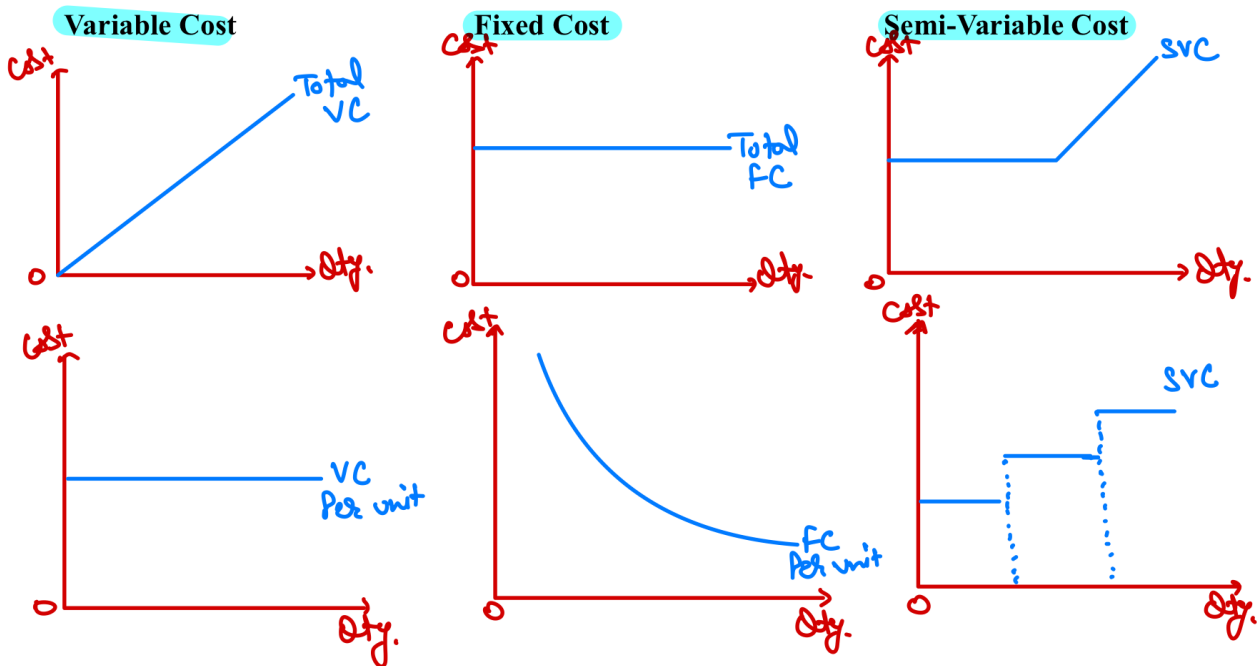
### 3. Types of budget



**4. Fixed Budget** – It is a budget designed to remain unchanged irrespective to the level of activity actually attained.

**5. Flexible Budget** – It is a budget which by recognizing the difference between fixed, semi-variable and variable costs is designed to change in relation to the level of activity attained.

## 6. Type of Costs



**7. Total Cost = No. of units × Cost per unit**

## 8. Quantity & Price Effect

|                 | Total Variable Cost | Total Fixed Cost |
|-----------------|---------------------|------------------|
| Quantity Effect | Yes                 | No               |
| Price Effect    | Yes                 | Yes              |

## 9. Points to Remember (PTRs)

Unless otherwise provided, following assumptions are to be taken

- VC per unit will remain same
- Total Fixed cost will remain same
- All direct cost are assumed to be variable
- All overheads are assumed to be fixed

$$\text{Wages} \propto \frac{1}{\text{Efficiency}} \quad \text{and} \quad \text{Efficiency} \propto \text{Output}$$

**10. Distribution of Semi-Variable Cost**

$$\text{Variable cost per unit out of SVC} = \frac{\text{Difference in cost}}{\text{Difference in units}}$$

$$\text{Fixed cost out of SVC} = \text{Total cost} - \text{Total variable cost}$$

**Question – 1**

PJ Ltd. manufactures hockey sticks. It sells the products at ₹ 500 each and makes a profit of ₹ 125 on each stick. The Company is producing 5,000 stocks annually by using 50% of its machinery capacity. The cost of each stick is as under:

|                  |                     |
|------------------|---------------------|
| Direct material  | ₹ 150               |
| Direct wages     | ₹ 50                |
| Work Overheads   | ₹ 125 (50% fixed)   |
| Selling Expenses | ₹ 50 (25% variable) |

The anticipation for the next year is that cost will go up as under:

|                 |     |
|-----------------|-----|
| Fixed charges   | 10% |
| Direct wages    | 20% |
| Direct material | 5%  |

There will not be any change in selling price. There is an additional order for 2,000 sticks in the next year. Calculate the lowest price that can be quoted so that the Company can earn the same profit as it earned in the current year?

**Solution****Statement of calculation of selling price**

| Particulars                                               | Amount (₹) |
|-----------------------------------------------------------|------------|
| Direct Material [(150 + 5%) × 7,000]                      | 11,02,500  |
| Direct Wages [(50 + 20%) × 7,000]                         | 4,20,000   |
| Variable Works Overhead [125 × 50% × 7,000]               | 4,37,500   |
| Fixed Works Overhead [125 × 50% × 5,000 × 110%]           | 3,43,750   |
| Variable Selling Expenses [50 × 25% × 7,000]              | 87,500     |
| Fixed Selling Expenses [50 × 75% × 5,000 × 110%]          | 2,06,250   |
| Total Cost                                                | 25,97,500  |
| Add: Desired Profit (125 × 7,000)                         | 8,75,000   |
| Total Sales Value                                         | 34,72,500  |
| Less: Existing Sales from 5,000 units [5,000 × 500]       | 25,00,000  |
| Sales value to be obtained from remaining 2,000 units (A) | 9,72,500   |
| Sale units (B)                                            | 2,000      |
| Selling price per unit (A ÷ B)                            | 486.25     |

**Question – 2**

During the FY 2020-21, SK Limited has produced 60,000 units operating at 50% capacity level. The cost structure at the 50% level of activity is as under:

|                                                  | (₹)          |
|--------------------------------------------------|--------------|
| Direct material                                  | 300 per unit |
| Direct wages                                     | 100 per unit |
| Variable overheads                               | 100 per unit |
| Direct Expenses                                  | 60 per unit  |
| Factory expenses (25% fixed)                     | 80 per unit  |
| Selling and Distribution expenses (80% variable) | 40 per unit  |
| Office and administrative expenses (100 % fixed) | 20 per unit  |

The company anticipates that in FY 2021-22, the variable costs will go up by 20% and fixed costs will go up by 15%. The selling price per unit will increase by 10% to ₹ 880. Required:

- Calculate the budgeted profit/loss for the FY 2020-21.
- Prepare an expense budget on marginal cost basis for the FY 2021-22 for the company at 50% and 60% level of activity and find out the profits at respective levels.

**Solution****(i) Calculation of Budgeted Profit for the year FY 2020-21**

|                                  | 60,000 Units |                  |
|----------------------------------|--------------|------------------|
|                                  | Per Unit     | Total            |
| Sales (A)                        | 800.00       | 4,80,00,000      |
| <b>Variable Cost</b>             |              |                  |
| Direct material                  | 300.00       | 1,80,00,000      |
| Direct wages                     | 100.00       | 60,00,000        |
| Variable overheads               | 100.00       | 60,00,000        |
| Direct expenses                  | 60.00        | 36,00,000        |
| Variable factory exp. (80×75%)   | 60.00        | 36,00,000        |
| Variable selling exp. (40×80%)   | 32.00        | 19,20,000        |
| Total Variable cost (B)          | 652          | 3,91,20,000      |
| <b>Fixed Cost</b>                |              |                  |
| Office and admin. Exp. (100%)    | -            | 12,00,000        |
| Fixed factory exp. (25%)         | -            | 12,00,000        |
| Fixed selling & dist. Exp. (20%) | -            | 4,80,000         |
| Total Fixed cost (C)             | -            | 28,80,000        |
| Total cost (B+C = D)             | -            | 4,20,00,000      |
| <b>Profit (A – D)</b>            | -            | <b>60,00,000</b> |

**(ii) Expenses Budget for the year FY 2021-22 at 50% & 60% level**

|           | 60,000 units |             | 72,000 units |             |
|-----------|--------------|-------------|--------------|-------------|
|           | Per Unit     | Total       | Per Unit     | Total       |
| Sales (A) | 880          | 5,28,00,000 | 880          | 6,33,60,000 |

|                                  |        |                  |        |                  |
|----------------------------------|--------|------------------|--------|------------------|
| <b>Variable Cost</b>             |        |                  |        |                  |
| Direct material                  | 360.00 | 2,16,00,000      | 360.00 | 2,59,20,000      |
| Direct wages                     | 120.00 | 72,00,000        | 120.00 | 86,40,000        |
| Variable overheads               | 120.00 | 72,00,000        | 120.00 | 86,40,000        |
| Direct expenses                  | 72.00  | 43,20,000        | 72.00  | 51,84,000        |
| Variable factory exp. (80×75%)   | 72.00  | 43,20,000        | 72.00  | 51,84,000        |
| Variable selling exp. (40×80%)   | 38.40  | 23,04,000        | 38.40  | 27,64,800        |
| Total Variable cost (B)          | 782.40 | 4,69,44,000      | 782.40 | 5,63,32,800      |
| <b>Fixed Cost</b>                |        |                  |        |                  |
| Office and admin. Exp. (100%)    | -      | 13,80,000        | -      | 13,80,000        |
| Fixed factory exp. (25%)         | -      | 13,80,000        | -      | 13,80,000        |
| Fixed selling & dist. Exp. (20%) | -      | 5,52,000         | -      | 5,52,000         |
| Total Fixed cost (C)             | -      | 33,12,000        | -      | 33,12,000        |
| Total cost (B + C = D)           | -      | 5,02,56,000      | -      | 5,96,44,800      |
| <b>Profit (A – D)</b>            | -      | <b>25,44,000</b> | -      | <b>37,15,200</b> |

**Question – 3**

The Accountant of KPMR Ltd. has prepared the following budget for the coming year 2022 for its two products 'AYE' and 'ZYE':

| Particulars                     | Product 'AYE'        | Product 'ZYE'        |
|---------------------------------|----------------------|----------------------|
| Production and Sales (in Units) | 4,000                | 3,000                |
|                                 | <b>Amount (in ₹)</b> | <b>Amount (in ₹)</b> |
| Selling price per unit          | 200                  | 180                  |
| Direct material per unit        | 80                   | 70                   |
| Direct labour per unit          | 40                   | 35                   |
| Variable overhead per unit      | 20                   | 25                   |
| Fixed overhead per unit         | 10                   | 10                   |

After reviewing the above budget, the management has called the marketing team for suggesting some measures for increasing the sales. The marketing team has suggested that by promoting the products on social media, the sales quantity of both the products can be increased by 5%. Also, the selling price per unit will go up by 10%. But this will result in increase in expenditure on variable overhead and fixed overhead by 20% and 5% respectively for both the products.

You are required to prepare flexible budget for both the products:

- Before promotion on social media
- After promotion on social media

**Solution****(i) Flexible Budget (Before promotion)**

| Particulars | Product AYE                   | Product ZYE                   | Total     |
|-------------|-------------------------------|-------------------------------|-----------|
| Sales       | $4,000 \times 200 = 8,00,000$ | $3,000 \times 180 = 5,40,000$ | 13,40,000 |

|                       |                              |                              |          |
|-----------------------|------------------------------|------------------------------|----------|
| Less: Direct Material | $4,000 \times 80 = 2,40,000$ | $3,000 \times 70 = 2,10,000$ | 4,50,000 |
| Less: Direct labour   | $4,000 \times 40 = 1,60,000$ | $3,000 \times 35 = 1,05,000$ | 2,65,000 |
| Less: Variable OHs    | $4,000 \times 20 = 80,000$   | $3,000 \times 25 = 75,000$   | 1,55,000 |
| Less: Fixed OHs       | $4,000 \times 10 = 40,000$   | $3,000 \times 10 = 30,000$   | 70,000   |
| Profit                | 2,80,000                     | 1,20,000                     | 4,00,000 |

**(ii) Flexible Budget (After promotion)**

| Particulars           | Product AYE                   | Product ZYE                   | Total     |
|-----------------------|-------------------------------|-------------------------------|-----------|
| Sales                 | $4,200 \times 220 = 9,24,000$ | $3,150 \times 198 = 6,23,700$ | 15,47,700 |
| Less: Direct Material | $4,200 \times 80 = 3,36,000$  | $3,150 \times 70 = 2,20,500$  | 5,56,500  |
| Less: Direct labour   | $4,200 \times 40 = 1,68,000$  | $3,150 \times 35 = 1,05,000$  | 2,73,000  |
| Less: Variable OHs    | $4,200 \times 24 = 1,00,800$  | $3,150 \times 25 = 1,10,250$  | 2,11,050  |
| Less: Fixed OHs       | $40,000 + 5\% = 42,000$       | $30,000 + 5\% = 31,500$       | 73,500    |
| Profit                | 2,77,200                      | 1,56,450                      | 4,33,650  |

**Question – 4**

SK Ltd. normally produce 8,000 units of their product in a month, in their machine shop. For the month of January, they had planned for a production of 10,000 units. Owing to a sudden cancellation of a contract in the middle of January, they could only produce 6,000 units in January.

Indirect manufacturing costs are carefully planned and monitored in the machine shop and the foreman of the shop is paid a 10% of the savings as bonus when in any month the indirect manufacturing cost incurred is less than the budgeted provision.

The foreman has put in a claim that he should be paid a bonus of ₹88.50 for the month of January. The works manager wonders how anyone can claim a bonus when the company has lost a sizeable contract. The relevant figures are as under:

| Indirect manufacturing  | Expenses for a normal month (₹) | Planned for January (₹) | Actual in costs January (₹) |
|-------------------------|---------------------------------|-------------------------|-----------------------------|
| Salary of foreman       | 1,000                           | 1,000                   | 1,000                       |
| Indirect labour         | 720                             | 900                     | 600                         |
| Indirect material       | 800                             | 1,000                   | 700                         |
| Repairs and maintenance | 600                             | 650                     | 600                         |
| Power                   | 800                             | 875                     | 740                         |
| Tools consumed          | 320                             | 400                     | 300                         |
| Rates and taxes         | 150                             | 150                     | 150                         |
| Depreciation            | 800                             | 800                     | 800                         |
| Insurance               | 100                             | 100                     | 100                         |
|                         | 5,290                           | 5,875                   | 4,990                       |

Do you agree with the works manager? Is the foreman entitled to any bonus for the performance in January? Substantiate your Answer with facts and figures. Explain.

### Solution

#### Flexible Budget (for the month of January)

| Indirect manufacturing                               | Nature of cost | Expenses for a normal month | Planned expenses for January | Expenses as per flexible budget for the month of January | Actual expenses of the month of January | Difference for Increased or (Decreased) |
|------------------------------------------------------|----------------|-----------------------------|------------------------------|----------------------------------------------------------|-----------------------------------------|-----------------------------------------|
|                                                      | (1)            | (₹)<br>(2)                  | (₹)<br>(3)                   | (₹)<br>(4)                                               | (₹)<br>(5)                              | (₹)<br>(6 = 5 - 4)                      |
| Salary of foreman                                    | Fixed          | 1,000                       | 1,000                        | 1,000                                                    | 1,000                                   | Nil                                     |
| Indirect labour<br>(Refer to working note 1)         | Variable       | 720                         | 900                          | 540                                                      | 600                                     | 60                                      |
| Indirect material<br>(Refer to working note 2)       | Variable       | 800                         | 1,000                        | 600                                                      | 700                                     | 100                                     |
| Repairs and maintenance<br>(Refer to working note 3) | Semi-variable  | 600                         | 650                          | 550                                                      | 600                                     | 50                                      |
| Power<br>(Refer to working note 4)                   | Semi-variable  | 800                         | 875                          | 725                                                      | 740                                     | 15                                      |
| Tools consumed<br>(Refer to working note 5)          | Variable       | 320                         | 400                          | 240                                                      | 300                                     | 60                                      |
| Rates and taxes                                      | Fixed          | 150                         | 150                          | 150                                                      | 150                                     | Nil                                     |
| Depreciation                                         | Fixed          | 800                         | 800                          | 800                                                      | 800                                     | Nil                                     |
| Insurance                                            | Fixed          | 100                         | 100                          | 100                                                      | 100                                     | Nil                                     |
| Total                                                |                | 5,290                       | 5,875                        | 4,705                                                    | 4,990                                   | 285                                     |

**Conclusion:** The above statement of flexible budget clearly shows that the concern's expenses in the month of January have increased from ₹4,705 to ₹4,990. Under such circumstances the Foreman of the company is not at all entitled for any performance bonus in January.

#### Working Notes:

1. Indirect labour cost per unit =  $\frac{720}{8,000} = ₹0.09$

Indirect labour for 6,000 units =  $6,000 \times 0.09 = ₹540$

2. Indirect material cost per unit =  $\frac{800}{8,000} = ₹0.10$

Indirect material for 6,000 units =  $6,000 \times 0.10 = ₹600$

3. According to high and low point method of segregating semi-variable cost into fixed and variable components, following formula may be used.

Variable cost of repair and maintenance per unit =  $\frac{\text{Change in expense level}}{\text{Change in Output level}} = \frac{650-600}{2,000} = ₹0.025$

For 8,000 units: Total variable cost of repair & maintenance =  $8,000 \times 0.025 = ₹200$

Fixed repair & maintenance cost =  $₹600 - 200 = ₹400$

For 6,000 units: Total cost of repair & maintenance =  $(6,000 \times 0.025) + 400 = ₹550$

4. Variable cost of power per unit =  $\frac{875-800}{2,000} = ₹0.0375$

For 8,000 units: Total variable cost of power =  $8,000 \times 0.0375 = ₹300$

Fixed repair & maintenance cost =  $₹800 - 300 = ₹500$

For 6,000 units: Total cost of repair & maintenance =  $(6,000 \times 0.0375) + 500 = ₹725$

5. Tools consumed cost for 6,000 units =  $\frac{320}{8,000} \times 6,000 = ₹240$

### Question – 5

The cost accountant of manufacturing company provides you the following details for year 2018:

|                            | ₹        |                     | ₹        |
|----------------------------|----------|---------------------|----------|
| Direct materials           | 1,75,000 | Other variable cost | 80,000   |
| Direct wages               | 1,00,000 | Other fixed costs   | 80,000   |
| Fixed factory overheads    | 1,00,000 | Profit              | 1,15,000 |
| Variable factory overheads | 1,00,000 | Sales               | 7,50,000 |

During the year, the company manufactured two products A and B and the output and costs were:

|                          | A        | B        |
|--------------------------|----------|----------|
| Output (units)           | 2,00,000 | 1,00,000 |
| Selling price per unit   | ₹2.00    | ₹3.50    |
| Direct material per unit | ₹0.50    | ₹0.75    |
| Direct wages per unit    | ₹0.25    | ₹0.50    |

Variable factory overheads are absorbed as a percentage of direct wages. Other variable costs have been computed as: Product A ₹0.25 per unit; and B ₹0.30 per unit.

During 2019, it is expected that the demand for product A will fall by 25% and for B by 50%. It is decided to manufacture a further product C, the cost for which are estimated as follows:

|                           | Product C |
|---------------------------|-----------|
| Output (units)            | 2,00,000  |
| Selling price per unit    | ₹1.75     |
| Direct materials per unit | ₹0.40     |
| Direct wages per unit     | ₹0.25     |

It is anticipated that the other variable costs per unit will be the same as for product A.

Prepare a budget to present to the management, showing the current position and the position for 2019. Comment on the comparative results.

### Solution

#### Budget Showing current Position and Position for 2019

| Particulars   | Current Position |            |          | Position for 2019 |          |              |          |
|---------------|------------------|------------|----------|-------------------|----------|--------------|----------|
|               | A                | B<br>(A+B) | Total    | A                 | B        | C<br>(A+B+C) | Total    |
| Sales (units) | 2,00,000         | 1,00,000   | -        | 1,50,000          | 50,000   | 2,00,000     | -        |
|               | ₹                | ₹          | ₹        | ₹                 | ₹        | ₹            | ₹        |
| A. Sales (₹)  | 4,00,000         | 3,50,000   | 7,50,000 | 3,00,000          | 1,75,000 | 3,50,000     | 8,25,000 |

| Particulars                  | Current Position |                 |                 | Position for 2019 |               |                 |                 |
|------------------------------|------------------|-----------------|-----------------|-------------------|---------------|-----------------|-----------------|
| Direct materials             | 1,00,000         | 75,000          | 1,75,000        | 75,000            | 37,500        | 80,000          | 1,92,500        |
| Direct wages                 | 50,000           | 50,000          | 1,00,000        | 37,500            | 25,000        | 50,000          | 1,12,500        |
| Factory overhead             | 50,000           | 50,000          | 1,00,000        | 37,500            | 25,000        | 50,000          | 1,12,500        |
| (var)                        | 50,000           | 30,000          | 80,000          | 37,500            | 15,000        | 50,000          | 1,02,500        |
| Other variable costs         | 2,50,000         | 2,05,000        | 4,55,000        | 1,87,500          | 10,25,000     | 2,30,000        | 5,20,000        |
| <b>B. Variable cost</b>      | <b>1,50,000</b>  | <b>1,45,000</b> | <b>2,95,000</b> | <b>1,12,500</b>   | <b>72,500</b> | <b>1,20,000</b> | <b>3,05,000</b> |
| <b>C. Contribution (A-B)</b> |                  |                 | 1,00,000        |                   |               |                 | 1,00,000        |
| Fixed cost: Factory          |                  |                 | 80,000          |                   |               |                 | 80,000          |
| Other                        |                  |                 | 1,80,000        |                   |               |                 | 1,80,000        |
| <b>D. Total fixed cost</b>   |                  |                 | <b>1,15,000</b> |                   |               |                 | <b>1,25,000</b> |
| <b>Profit (C – D)</b>        |                  |                 |                 |                   |               |                 |                 |

**Comments:** Introduction of product C is likely to increase profit by ₹10,000 (i.e. from ₹1,15,000 to ₹1,25,000) in 2019 as compared to 2018 Therefore, introduction of product C is recommended.

### 11. Sales Budget

| Product      | Units    | Selling price per unit | Sales    |
|--------------|----------|------------------------|----------|
| A            | -        | -                      | -        |
| B            | -        | -                      | -        |
| C            | -        | -                      | -        |
| <b>Total</b> | <b>-</b> |                        | <b>-</b> |

### 12. Production Budget

| Particulars             | Product A | Product B | Product C |
|-------------------------|-----------|-----------|-----------|
| Sales                   | -         | -         | -         |
| (+) Closing Stock       | -         | -         | -         |
| (-) Opening Stock       | -         | -         | -         |
| <b>Production units</b> | <b>-</b>  | <b>-</b>  | <b>-</b>  |

### 13. Raw Material Consumption Budget

| Product | Prod. Quantity | RM consumption per unit | Total RM Consumption |
|---------|----------------|-------------------------|----------------------|
| A       | -              | -                       | -                    |
| B       | -              | -                       | -                    |
| C       | -              | -                       | -                    |
|         | -              | -                       | -                    |

### 14. Raw Material Purchase Budget

| Particulars                   | January  | February | March    |
|-------------------------------|----------|----------|----------|
| Raw Material Consumed         | -        | -        | -        |
| (+) Closing Stock             | -        | -        | -        |
| (-) Opening Stock             | -        | -        | -        |
| <b>Raw Material Purchased</b> | <b>-</b> | <b>-</b> | <b>-</b> |

**15. Labour Budget**

| Particulars                         | Product A | Product B | Product C |
|-------------------------------------|-----------|-----------|-----------|
| Production Units                    | -         | -         | -         |
| Direct labour hour per unit         | -         | -         | -         |
| Total Labour hours                  | -         | -         | -         |
| Total Labour cost @ Rs. __ per hour | -         | -         | -         |

**Question – 6**

PSV Ltd. manufactures and sells a single product and estimated the following related information for the period November, 2020 to March, 2021.

| Particulars                                | November, 2020 | December, 2020 | January, 2021 | February, 2021 | March, 2021 |
|--------------------------------------------|----------------|----------------|---------------|----------------|-------------|
| Opening Stock of Finished goods (in Units) | 7,500          | 3,000          | 9,000         | 8,000          | 6,000       |
| Sales (in Units)                           | 30,000         | 35,000         | 38,000        | 25,000         | 40,000      |
| Selling Price per unit (in ₹)              | 10             | 12             | 15            | 15             | 20          |

**Additional information:**

- Closing stock of finished goods at the end of march, 2021 is 10,000 units
- Each unit of finished output requires 2kg of Raw Material 'A' and 3kg of Raw Material 'B'.

You are required to prepare the following budgets for the period November, 2020 to March 2021 on monthly basis:

- Sales budget (in ₹)
- Production Budget (in units) and
- Raw material budget for raw material 'A' and 'B' separately (in units)

**Solution****(i) Sales Budget**

| Particulars                   | November, 2020 | December, 2020 | January, 2021 | February, 2021 | March, 2021 |
|-------------------------------|----------------|----------------|---------------|----------------|-------------|
| Sales (in Units)              | 30,000         | 35,000         | 38,000        | 25,000         | 40,000      |
| Selling Price per unit (in ₹) | 10             | 12             | 15            | 15             | 20          |
| Sales Value                   | 3,00,000       | 4,20,000       | 5,70,000      | 3,75,000       | 8,00,000    |

**(ii) Production Budget**

| Particulars               | November, 2020 | December, 2020 | January, 2021 | February, 2021 | March, 2021 |
|---------------------------|----------------|----------------|---------------|----------------|-------------|
| Sales Units               | 30,000         | 35,000         | 38,000        | 25,000         | 40,000      |
| Add: Closing Stock Units  | 3,000          | 9,000          | 8,000         | 6,000          | 10,000      |
| Less: Opening Stock Units | (7,500)        | (3,000)        | (9,000)       | (8,000)        | (6,000)     |
| Production Units          | 25,500         | 41,000         | 37,000        | 23,000         | 44,000      |

(iii)

## Raw Material 'A' Budget

| Particulars                       | November, 2020 | December, 2020 | January, 2021 | February, 2021 | March, 2021 |
|-----------------------------------|----------------|----------------|---------------|----------------|-------------|
| Production Units                  | 25,500         | 41,000         | 37,000        | 23,000         | 44,000      |
| Raw material consumption per unit | 2              | 2              | 2             | 2              | 2           |
| Raw Material Consumption          | 51,000         | 82,000         | 74,000        | 46,000         | 88,000      |

## Raw Material 'B' Budget

| Particulars                       | November, 2020 | December, 2020 | January, 2021 | February, 2021 | March, 2021 |
|-----------------------------------|----------------|----------------|---------------|----------------|-------------|
| Production Units                  | 25,500         | 41,000         | 37,000        | 23,000         | 44,000      |
| Raw material consumption per unit | 3              | 3              | 3             | 3              | 3           |
| Raw Material Consumption          | 76,500         | 1,23,000       | 1,11,000      | 69,000         | 1,32,000    |

## Question – 7

SK Ltd. is drawing a production plan for its two products Minimax (MM) and Heavyhigh (HH) for the year 2019-20. The company's policy is to hold closing stock of finished goods at 25% of the anticipated volume of sales of the succeeding month. The following are the estimated data for two products:

|                               | Minimax (MM) | Heavyhigh (HH) |
|-------------------------------|--------------|----------------|
| Budgeted Production units     | 1,80,000     | 1,20,000       |
|                               | (₹)          | (₹)            |
| Direct material cost per unit | 220          | 280            |
| Direct labour cost per unit   | 130          | 120            |
| Manufacturing overhead        | 4,00,000     | 5,00,000       |

The estimated units to be sold in the first four months of the year 2019-20 are as under:

|           | April | May    | June   | July   |
|-----------|-------|--------|--------|--------|
| Minimax   | 8,000 | 10,000 | 12,000 | 16,000 |
| Heavyhigh | 6,000 | 8,000  | 9,000  | 14,000 |

Prepare production budget for the first quarter in month-wise.

**Solution**

## Production budget (in Units)

|                     | April   |         | May     |         | June    |         | Total   |         |
|---------------------|---------|---------|---------|---------|---------|---------|---------|---------|
|                     | MM      | HH      | MM      | HH      | MM      | HH      | MM      | HH      |
| Sales               | 8,000   | 6,000   | 10,000  | 8,000   | 12,000  | 9,000   | 30,000  | 23,000  |
| Add: Closing stock  | 2,500   | 2,000   | 3,000   | 2,250   | 4,000   | 3,500   | 9,500   | 7,750   |
| Less: Opening stock | (2,000) | (1,500) | (2,500) | (2,000) | (3,000) | (2,250) | (7,500) | (5,750) |
| Production units    | 8,500   | 6,500   | 10,500  | 8,250   | 13,000  | 10,250  | 32,000  | 25,000  |

## Production Cost Budget

| Element of cost                                                                                | Rate (₹)          |                   | Amount (₹)  |             |
|------------------------------------------------------------------------------------------------|-------------------|-------------------|-------------|-------------|
|                                                                                                | MM (32,000 units) | HH (25,000 units) | MM          | HH          |
| Direct Material                                                                                | 220               | 280               | 70,40,000   | 70,00,000   |
| Direct Labour                                                                                  | 130               | 120               | 41,60,000   | 30,00,000   |
| Manufacturing Overhead<br>[(4,00,000 ÷ 1,80,000) × 32,000]<br>[(5,00,000 ÷ 1,20,000) × 25,000] |                   |                   | 71,111      | 1,04,167    |
|                                                                                                |                   |                   | 1,12,71,111 | 1,01,04,167 |

## Question – 8

AB manufacturing Company manufactures two products A and B. Both Products use a common Raw Material 'C'. The Raw Material 'C' is purchased at the rate of ₹ 45 per kg from the Market. The Company has made estimates for the year ended 31<sup>st</sup> March, 2018 (the budget period) as under:

|                                                      | Product A | Product B |
|------------------------------------------------------|-----------|-----------|
| Sales in Units                                       | 36,000    | 16,700    |
| Finished goods stock increase by year-end (in Units) | 860       | 400       |
| Post-production Rejection Rate (%)                   | 3         | 5         |
| Material 'C' per completed Unit, net of wastage      | 4 kg      | 5 kg      |
| Material 'C' wastage in %                            | 5         | 4         |

Additional information available is as under:

- Usage of Raw Material 'C' is expected to be at a constant rate over the period.
- Annual cost of holding one unit of Raw Material 'C' in Stock is 9% of the Material Cost.
- The cost of placing an order is ₹ 250 per order.

You are required to:

- Prepare Functional Budgets for the year ended 31<sup>st</sup> March, 2018 under the following categories:
  - Production Budget for Products A and B in Units
  - Purchase Budget for Raw Material 'C' in kg and value.
- Calculate the Economic Order Quantity (EOQ) in kg for Raw Material 'C'.

**Solution**

- Production Budget (in units) for the year ended 31<sup>st</sup> March 2018**

| Particulars                      | Product A             | Product B             |
|----------------------------------|-----------------------|-----------------------|
| Budgeted sales (units)           | 36,000                | 16,700                |
| Add: Increase in closing stock   | 860                   | 400                   |
| No. of good units to be produced | 36,860                | 17,100                |
| Post production rejection rate   | 3%                    | 5%                    |
| Post production good units rate  | 100% – 3% = 97%       | 100% - 5% = 95%       |
| No. of units to be produced      | 36,860 ÷ 97% = 38,000 | 17,100 ÷ 95% = 18,000 |

## (ii) Purchase budget (in kgs and value) for Material C

| Particulars                                | Product A                        | Product B                      |
|--------------------------------------------|----------------------------------|--------------------------------|
| No. of units to be produced                | 38,000                           | 18,000                         |
| Usage of Material C per unit of production | 4 kg                             | 5 kg                           |
| Material needed for production             | 1,52,000 kg                      | 90,000 kg                      |
| Wastage % of Material C                    | 5%                               | 4%                             |
| Good usage % of Material C                 | 100% - 5% = 95%                  | 100% - 4% = 96%                |
| Material to be purchased (in kg)           | $1,52,000 \div 95\% = 1,60,000$  | $90,000 \div 96\% = 93,750$    |
| Rate per kg of Material C                  | ₹ 45                             | ₹ 45                           |
| Total Purchase cost                        | $1,60,000 \times 45 = 72,00,000$ | $93,750 \times 45 = 42,18,750$ |

Total purchase cost = 72,00,000 + 42,18,750 = ₹ 1,14,18,750

(iii)  $A = 1,60,000 + 93,750 = 2,53,750$  kg

$$O = ₹ 250$$

$$C = ₹ 45 \times 9\% = ₹ 4.05$$

$$EOQ = \sqrt{\frac{2 \times A \times O}{C}} = \sqrt{\frac{2 \times 2,53,750 \times 250}{4.05}} = 5,597 \text{ kg}$$

**Question – 9**

A Limited has furnished the following information for the months from 1<sup>st</sup> January to 30<sup>th</sup> April, 2023:

|                                                            | January | February | March | April |
|------------------------------------------------------------|---------|----------|-------|-------|
| Number of working days                                     | 25      | 24       | 26    | 25    |
| Production (in units) per working day                      | 50      | 55       | 60    | 52    |
| Raw material purchases (% by weights to total of 4 months) | 21%     | 26%      | 30%   | 23%   |
| Purchase price of raw material (per kg)                    | ₹ 10    | ₹ 12     | ₹ 13  | ₹ 11  |

Quantity of raw material per unit of product: 4 kg

Opening stock of raw material on 1<sup>st</sup> January: 6,020 kg. (Cost ₹ 63,210)

Closing stock of raw material on 30<sup>th</sup> April: 5,100 kg.

All the purchases of material are made at the start of each month.

Required:

- Calculate the consumption of raw material (in kgs) month-by-month and in total
- Calculate the month-wise quantity and value of raw materials purchased
- Prepare the priced stores ledger for each month using the FIFO method.

**Solution**

## (i) Calculation of consumption for Raw Material (in kgs) month by month and total

| Particulars                    | Jan   | Feb   | March | April | Total  |
|--------------------------------|-------|-------|-------|-------|--------|
| No. of working days            | 25    | 24    | 26    | 25    | -      |
| Production (per day)           | 50    | 55    | 60    | 52    | -      |
| Production                     | 1,250 | 1,320 | 1,560 | 1,300 | 5,430  |
| Raw material consumed (in kgs) | 5,000 | 5,280 | 6,240 | 5,200 | 21,720 |

**Calculation of Raw Material Purchased**

| <b>Purchased</b>                               | <b>Kg</b> |
|------------------------------------------------|-----------|
| Closing stock on 30 <sup>th</sup> April        | 5,100     |
| Add: Raw material consumed                     | 21,720    |
| Less: Opening stock on 1 <sup>st</sup> January | (6,020)   |
| Raw material purchased                         | 20,800    |

**(ii) Calculation of month wise quantity and value of raw materials purchased**

| <b>Month</b> | <b>Purchase quantity (Kgs)</b> | <b>Price (₹)</b> | <b>Value (₹)</b> |
|--------------|--------------------------------|------------------|------------------|
| January      | $20,800 \times 21\% = 4,368$   | 10               | 43,680           |
| February     | $20,800 \times 26\% = 5,408$   | 12               | 64,896           |
| March        | $20,800 \times 30\% = 6,240$   | 13               | 81,120           |
| April        | $20,800 \times 23\% = 4,784$   | 11               | 52,624           |
| Total        | 20,800                         |                  | 2,42,320         |

**(iii) Stores Price Ledger by using FIFO Method**

| <b>Month</b> | <b>Receipts</b>  |                 |               | <b>Issues</b>    |                 |               | <b>Balance</b>   |                 |               |
|--------------|------------------|-----------------|---------------|------------------|-----------------|---------------|------------------|-----------------|---------------|
|              | <b>Qty. (kg)</b> | <b>Rate (₹)</b> | <b>Amount</b> | <b>Qty. (kg)</b> | <b>Rate (₹)</b> | <b>Amount</b> | <b>Qty. (kg)</b> | <b>Rate (₹)</b> | <b>Amount</b> |
| Jan          | -                | -               | -             | -                | -               | -             | 6,020            | 10.5            | 63,210        |
| Jan          | 4,368            | 10              | 43,680        | -                | -               | -             | 6,020            | 10.5            | 63,210        |
|              |                  |                 |               |                  |                 |               | 4,368            | 10              | 43,680        |
| Jan          | -                | -               | -             | 5,000            | 10.5            | 52,500        | 1,020            | 10.5            | 10,710        |
|              |                  |                 |               |                  |                 |               | 4,368            | 10              | 43,680        |
| Feb          | 5,408            | 12              | 64,896        | -                | -               | -             | 1,020            | 10.5            | 10,710        |
|              |                  |                 |               |                  |                 |               | 4,368            | 10              | 43,680        |
|              |                  |                 |               |                  |                 |               | 5,408            | 12              | 64,896        |
| Feb          | -                | -               | -             | 1,020            | 10.5            | 10,710        | 108              | 10              | 1,080         |
|              |                  |                 |               | 4,260            | 10              | 42,600        | 5,408            | 12              | 64,896        |
| March        | 6,240            | 13              | 81,120        | -                | -               | -             | 108              | 10              | 1,080         |
|              |                  |                 |               |                  |                 |               | 5,408            | 12              | 64,896        |
|              |                  |                 |               |                  |                 |               | 6,240            | 13              | 81,120        |
| March        | -                | -               | -             | 108              | 10              | 1,080         | -                | -               | -             |
|              |                  |                 |               | 5,408            | 12              | 64,896        | 5,516            | 13              | 71,708        |
|              |                  |                 |               | 724              | 13              | 9,412         |                  |                 |               |
| April        | 4,784            | 11              | 52,624        | -                | -               | -             | 5,516            | 13              | 52,624        |
|              |                  |                 |               |                  |                 |               | 4,784            | 11              | 71,708        |
| April        | -                | -               | -             | 5,200            | 13              | 67,600        | 316              | 13              | 4,108         |
|              |                  |                 |               |                  |                 |               | 4,784            | 11              | 52,624        |

**Question – 10**

SK Ltd. Produces and sells a single product. Sales budget for the calendar year 2019 by quarter is as under:

| Quarter | No. of units to be sold |
|---------|-------------------------|
| I       | 12,000                  |
| II      | 15,000                  |
| III     | 16,500                  |
| IV      | 18,000                  |

The year is expected to open with an inventory of 4,000 units of finished product and close with an inventory of 6,500 units.

Production is customarily scheduled to provide for two-thirds of the current quarter's sales demand plus one third of the following quarter's demand. Thus, production anticipates sales volume by about one month.

The standard cost details for one unit of the product is as follows:

Direct materials 10 lbs. @ 50 paise per lb.

Direct labour 1 hour 30 minutes @ ₹ 4 per hour.

Variable overheads 1 hour 30 minutes @ ₹ 1 per hour.

Fixed overheads 1 hour 30 minutes @ ₹ 2 per hour based on a budgeted volume of 90,000 direct labour hour for the year.

- Prepare a production budget for 2019, by quarters, showing the number of units to be produced, and the total costs of direct labour, variable overheads and fixed overheads.
- If the budgeted selling price per unit is ₹ 17, what would be the budgeted profit for the year as a whole?
- In which quarter of the year is the company expected to breakeven?

**Solution****(i) Production Budget**

| Particulars          | Quarter - 1   | Quarter – 2   | Quarter – 3   | Quarter – 4   | Total         |
|----------------------|---------------|---------------|---------------|---------------|---------------|
| Sales units          | 12,000        | 15,000        | 16,500        | 18,000        | 61,500        |
| <b>Production</b>    |               |               |               |               |               |
| 2/3 of current month | 8,000         | 10,000        | 11,000        | 12,000        | 41,000        |
| 1/3 of next month    | 5,000         | 5,500         | 6,000         | 6,500*        | 23,000        |
| <b>Production</b>    | <b>13,000</b> | <b>15,500</b> | <b>17,000</b> | <b>18,500</b> | <b>64,000</b> |

\*This value of 6,500 units is computed as balancing figure.

**Working Note -**

Annual total production = Sales + closing stock – Opening stock = 61,500 + 6,500 – 4,000 = 64,000

Production for Quarter - 4 = Total production – production of first three quarters  
 = 64,000 – 13,000 – 15,500 – 17,000 = 18,500

## Statement of Cost

| Particulars                                 | Quarter - 1 | Quarter - 2 | Quarter - 3 | Quarter - 4 | Total                    |
|---------------------------------------------|-------------|-------------|-------------|-------------|--------------------------|
| Production units                            | 13,000      | 15,500      | 17,000      | 18,500      | 64,000                   |
| Direct material ( $10 \times 0.5 = 5$ p.u.) | 65,000      | 77,500      | 85,000      | 92,500      | 3,20,000                 |
| Direct labour ( $1.5 \times 4 = 6$ p.u.)    | 78,000      | 93,000      | 1,02,000    | 1,11,000    | 3,84,000                 |
| Variable Ohs ( $1.5 \times 1 = 1.5$ p.u.)   | 19,500      | 23,250      | 25,500      | 27,750      | 96,000                   |
| Fixed overheads                             | 45,000      | 45,000      | 45,000      | 45,000      | 1,80,000<br>(90,000 × 2) |

(ii)

## Statement of profit

| Particulars                                                                                      | Amount    |
|--------------------------------------------------------------------------------------------------|-----------|
| Sales ( $17 \times 61,500$ )                                                                     | 10,45,500 |
| Less: Variable cost $\left[ \frac{(3,20,000 + 3,84,000 + 96,000)}{64,000} \times 61,500 \right]$ | 7,68,750  |
| <b>Contribution</b>                                                                              | 2,76,750  |
| Less: Fixed costs                                                                                | 1,80,000  |
| <b>Profit</b>                                                                                    | 96,750    |

(iii) Breakeven point =  $\frac{\text{Fixed Cost}}{\text{Contribution per unit}} = \frac{1,80,000}{\left( \frac{2,76,750}{61,500} \right)} = 40,000$  units

| Particulars            | Quarter - 1 | Quarter - 2 | Quarter - 3 | Quarter - 4 |
|------------------------|-------------|-------------|-------------|-------------|
| Sales units            | 12,000      | 15,000      | 16,500      | 18,000      |
| Cumulative Sales units | 12,000      | 27,000      | 43,500      | 61,500      |

∴ Breakeven will be achieved in quarter 3.

## Question – 11

SK Ltd. manufactures two products using two types of materials and one grade of labour. Shown below is an extract from the company's working papers for the next month's budget:

|                                                   | Product-A | Product-B |
|---------------------------------------------------|-----------|-----------|
| Budgeted sales (in units)                         | 2,400     | 3,600     |
| Budgeted material consumption per unit (in kg):   |           |           |
| Material-X                                        | 5         | 3         |
| Material-Y                                        | 4         | 6         |
| Standard labour hours allowed per unit of product | 3         | 5         |

Material-X and Material-Y cost ₹ 4 and ₹ 6 per kg and labours are paid ₹ 25 per hour. Overtime premium is 50% and is payable, if a worker works for more than 40 hours a week. There are 180 direct workers.

The target productivity ratio (or efficiency ratio) for the productive hours worked by the direct workers in actually manufacturing the products is 80%. In addition the non-productive down-time is budgeted at 20% of the productive hours worked.

There are four 5-days weeks in the budgeted period and it is anticipated that sales and production will occur evenly throughout the whole period.

It is anticipated that stock at the beginning of the period will be:

Product-A = 400 units; Product-B = 200 units;

Material-X = 1,000 kgs; Material-Y = 500 kgs.

The anticipated closing stocks for budget period are as below:

Product-A 4 days sales

Product-B 5 days sales

Material-X 10 days consumption

Material-Y 6 days consumption

Required to calculate the Material Purchase Budget and the Wages Budget for the direct workers, showing the quantities and values, for the next month.

### Solution

Number of days in budget period = 4 weeks × 5 days = 20 days

#### Number of units to be produced

|                                                                                                                                                                                    | Product-A<br>(units) | Product-B<br>(units) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------|
| Budgeted Sales                                                                                                                                                                     | 2,400                | 3,600                |
| Add: Closing stock $\left(\frac{2,400 \text{ units}}{20 \text{ days}} \times 4 \text{ days}\right) \left(\frac{3,600 \text{ units}}{20 \text{ days}} \times 5 \text{ days}\right)$ | 480                  | 900                  |
| Less: Opening stock                                                                                                                                                                | 400                  | 200                  |
|                                                                                                                                                                                    | 2,480                | 4,300                |

#### (i) Material purchase budget

|                                                                                                                                                                                       | Material-X (kg)                | Material-Y (kg)                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------|
| Material required:                                                                                                                                                                    |                                |                                |
| Product-A                                                                                                                                                                             | 12,400<br>(2,480 units x 5 kg) | 9,920<br>(2,480 units x 4 kg)  |
| Product-B                                                                                                                                                                             | 12,900<br>(4,300 units x 3 kg) | 25,800<br>(4,300 units x 6 kg) |
|                                                                                                                                                                                       | 25,300                         | 35,720                         |
| Add: Closing Stock<br>$\left(\frac{25,300 \text{ kg}}{20 \text{ days}} \times 10 \text{ days}\right) \left(\frac{35,720 \text{ units}}{20 \text{ days}} \times 6 \text{ days}\right)$ | 12,650                         | 10,716                         |
| Less: Opening stock                                                                                                                                                                   | 1,000                          | 500                            |
| Quantity to be purchased                                                                                                                                                              | 36,950                         | 45,936                         |
| Rate per kg. of material                                                                                                                                                              | ₹ 4                            | ₹ 6                            |
| <b>Total cost</b>                                                                                                                                                                     | <b>₹ 1,47,800</b>              | <b>₹ 2,75,616</b>              |

**(ii) Wages Budget**

|                                          | Product-A (Hours)                          | Product-B (Hours)                            |
|------------------------------------------|--------------------------------------------|----------------------------------------------|
| Units to be produced                     | 2,480 units                                | 4,300 units                                  |
| Standard hours allowed per unit          | 3                                          | 5                                            |
| Total standard hours allowed             | 7,440                                      | 21,500                                       |
| Productive hours required for production | $\frac{7,440 \text{ hours}}{80\%} = 9,300$ | $\frac{21,500 \text{ hours}}{80\%} = 26,875$ |
| Add: Non-Productive down time            | 1,860 hours<br>(20% of 9,300 hours)        | 5,375 hours<br>(20% of 26,875 hours)         |
| Hours to be paid                         | 11,160                                     | 32,250                                       |

Total Hours to be paid = 43,410 hours (11,160 + 32,250)

Hours to be paid at normal rate = 4 weeks × 40 hours × 180 workers = 28,800 hours

Hours to be paid at premium rate = 43,410 hours – 28,800 hours = 14,610 hours

Total wages to be paid = (28,800 hours × ₹ 25) + (14,610 hours × ₹ 37.5)  
= ₹ 7,20,000 + ₹ 5,47,875 = ₹ 12,67,875

**16. Zero Base Budgeting –** It is defined as ‘a method of budgeting which requires each cost element to be specifically justified, although the activities to which the budget relates are being undertaken for the first time, without approval, the budget allowance is zero’.

ZBB is an activity based budgeting system where budgets are prepared for each activities rather than functional department. Justification in the form of cost benefits for the activity is required to be given. The activities are then evaluated and prioritized by the management on the basis of factors like synchronisation with organisational objectives, availability of funds, regulatory requirement etc.

**17. Performance Budgeting –** It involves evaluation of the performance of an organisation in the context of both specific as well as overall objectives of the organisation. Performance Budgeting provide a meaningful relationship between estimated inputs and expected outputs as an integral part of the budgeting system. A performance budget is one which presents the purposes and objectives for which funds are required, the costs of the programmes proposed for achieving those objectives, and quantitative data measuring the accomplishments and work performed under each programme. Thus, PB is a technique of presenting budgets for costs and revenues in terms of functions. Programmes and activities correlate the physical and financial aspect of the individual items comprising the budget.